



EduTech Academy

Startup Manual for an Online and Brick-and-Mortar STEM Learning Ecosystem

YouTube -> Free Tutorials -> CSEC Subscription -> Robotics Camp -> Starter Kit -> STEM Academy -> Advanced Kit -> Competition Team

Prepared as a practical step-by-step operating guide for launching, validating, and scaling EduTech Academy. Version 1.0 | June 2026

1. Business System Overview

EduTech Academy should be built as a staged ecosystem, not a single course website. Each offer has a job: attract attention, create trust, generate recurring revenue, deepen student engagement, and eventually turn top learners into high-visibility STEM performers.

Stage	Main Offer	Business Purpose	Next Conversion
1. YouTube	Short free tutorials, exam tips, problem walkthroughs, robotics demos	Build trust, prove teaching quality, collect audience questions, establish brand authority	Free website visit and email lead
2. Free Tutorials	Selected lessons, worksheets, study guides, quiz samples	Give genuine value and show the structured learning path	CSEC subscription trial or paid plan
3. CSEC Subscription	Course pages, videos, quizzes, live revision classes, printed guide requests	Create recurring revenue and student outcomes in core exam subjects	Robotics camp enrolment
4. Robotics Camp	Holiday or weekend STEM camp with hands-on building and coding	Create premium experiences and parent excitement	Starter Kit purchase
5. Starter Kit	Beginner robotics/electronics kit plus guided lessons	Extend learning at home and increase product revenue	STEM Academy membership
6. STEM Academy	Physical classes, labs, clubs, certifications, mentorship	Become the local STEM hub and deepen retention	Advanced Kit and specialist tracks
7. Advanced Kit	Sensors, microcontrollers, motors, AI/IoT modules, project briefs	Move students from activity to serious engineering practice	Competition team selection
8. Competition Team	Elite training, competitions, showcases, sponsorships	Create reputation, student prestige, and partner visibility	New YouTube stories and recruitment loop

Operating principle: do not launch all stages at once. Start with the smallest offer that proves demand, then let student success and parent confidence fund the next layer.

Core Promise

- For students: clear STEM explanations, practice, feedback, confidence, and visible progress.
- For parents: trustworthy academic support, structured learning, safety, and future-ready skills.
- For schools and sponsors: measurable STEM engagement, talent development, and community impact.
- For EduTech Academy: recurring subscription revenue supported by high-value camps, kits, classes, and competition programs.

Minimum Viable Ecosystem

- 1 Publish free YouTube tutorials and direct every viewer to the website.
- 2 Offer a free lesson page with video, notes, quiz, and email capture.

- 3 Launch one paid CSEC subject subscription, beginning with Biology or Mathematics.
- 4 Run one small robotics workshop or camp as a pilot.
- 5 Package a starter kit for camp students and early adopters.
- 6 Use pilot results to decide whether to expand into a permanent STEM Academy space.

2. Foundation Setup Before Selling

Complete these setup tasks before accepting paid students. The goal is to look credible, operate safely, and avoid rebuilding basic systems after launch.

Area	Required Setup	Owner Check
Legal and admin	Register the business, choose the trading name, open a business bank account, set bookkeeping categories, and confirm local requirements for classes involving minors.	Do not treat this manual as legal advice. Verify requirements with local authorities and a qualified advisor.
Brand	Use one logo, one color palette, one promise, and one clear domain. Reserve social handles early.	Brand appears consistently on website, videos, worksheets, invoices, and kit labels.
Child safety	Create parent consent forms, attendance logs, pickup rules, emergency contacts, photo/video permissions, and incident reporting forms.	Every physical class has a signed consent and emergency contact record.
Payments	Set up subscription payment processing, receipt emails, cancellation rules, refund policy, and manual payment backup.	Test a payment from start to receipt before public launch.
Data	Keep student records, quiz scores, parent contacts, and payment data in controlled systems only.	Never store sensitive student data in public GitHub files.

Brand Positioning

Position EduTech Academy as the bridge between exam success and practical STEM capability. The message should be simple: students get better grades while learning the technology skills that make STEM feel real.

- Tagline option: Exam-ready. Future-ready.
- Audience: CSEC and CAPE students, parents, schools, and community sponsors.
- Tone: disciplined, encouraging, practical, modern, and Caribbean-relevant.
- Proof assets: sample lessons, student results, project photos, camp videos, testimonials, and competition outcomes.

3. Stage 1: YouTube and Free Tutorials

YouTube is the top of the funnel. Its job is not to hold the entire business. Its job is to attract students, answer common questions, and send serious learners to the EduTech Academy website.

YouTube Setup Steps

- 1 Create or polish the EduTech Academy YouTube channel with banner, logo, description, and website link.
- 2 Create playlists by audience: CSEC Biology, CSEC Mathematics, Robotics Basics, Exam Strategy, and Parent Guides.
- 3 Record 10 short videos before public promotion so the channel feels active.
- 4 End every video with one call to action: visit the website for notes, quizzes, and the next lesson.
- 5 Pin a comment under each video linking to the matching lesson page.
- 6 Track weekly views, click-throughs, comments, and subscriber growth.

Free Tutorial Content Formula

Video Type	Length	Purpose	Call to Action
Concept explainer	5-8 min	Teach one syllabus objective clearly	Download the free summary guide
Past paper walkthrough	8-15 min	Show exam reasoning and marking language	Try the self-assessment quiz
Common mistake video	3-6 min	Build trust by fixing pain points	Join the full revision path
Robotics demo	3-7 min	Create excitement for camps and kits	Join the camp waiting list
Parent guide	4-6 min	Explain offer, safety, outcomes, and timetable	Book a consultation or class

Publishing Rhythm

- Two academic videos per week while building the CSEC subscription.
- One short robotics or STEM demo per week during camp promotion periods.
- One live or premiere-style revision session before major exam windows.
- One monthly parent-facing video explaining progress, study discipline, or STEM careers.

4. Stage 2: CSEC Subscription

The subscription converts attention into recurring revenue. Start with one subject and make the experience excellent before expanding. The first subject should include videos, notes, quizzes, live revision options, and a way to request printed guides.

Subscription Build Steps

- 1 Choose the first subject and map the syllabus into sections, lessons, quizzes, and revision checkpoints.
- 2 Create a free preview path with two or three lessons available without payment.
- 3 Place paid lessons behind a subscription system or private course platform.
- 4 Create weekly live revision classes using Google Meet, Zoom, or another reliable platform.
- 5 Add self-assessment quizzes after each lesson and a cumulative quiz after each section.
- 6 Create a printed guide request form with delivery/pickup options and pricing.
- 7 Set up onboarding emails: welcome, how to study, how to join live class, how to request help.

Recommended Course Structure

Component	Free Tier	Paid Tier
Videos	A few selected concept videos	Full lesson sequence and exam walkthroughs
Notes	Summarised guide for sample topics	Complete section guides and printable study packs
Quizzes	Sample quiz with instant feedback	Full lesson quizzes, section tests, and score tracking
Live classes	Occasional public revision session	Scheduled subscriber revision classes
Support	Public comments or contact form	Priority question submission and class booking
AI tools	Public study tips	NotebookLM/Gemini-supported summaries, question banks, and revision prompts

Platform Decision

Use the static website and GitHub/Netlify for public pages, but do not force the static site to do everything. Use specialist tools for payments, user accounts, email, and video hosting as the business grows.

- Start simple: GitHub Pages or Netlify for public pages, YouTube for free videos, forms for enquiries, and manual enrolment if volume is low.
- When recurring payments begin: use Stripe or a local payment provider where available, plus a member/course system.
- When course complexity increases: consider Moodle, Tutor LMS/LearnDash, or a custom student portal.
- For paid videos: use a proper video platform such as Bunny Stream, Vimeo Pro, Cloudflare Stream, or a course platform with protected video delivery.

5. Stage 3: Robotics Camp

The robotics camp is the first premium hands-on product. It should be exciting, structured, safe, and easy for parents to understand. Start with a small pilot before renting large venues or buying excess equipment.

Pilot Camp Model

Decision	Recommended Pilot
Age group	10-16, divided into beginner and intermediate groups if numbers allow
Class size	8-16 students for the first camp
Duration	3-5 days during school vacation, or 4 Saturdays
Outcome	Every student builds and presents a working robot or STEM project
Instructor ratio	1 lead instructor plus 1 assistant for every 10-12 students
Parent moment	Final-day showcase with photos, certificates, and upgrade offer

Camp Setup Checklist

- Confirm venue, tables, chairs, power strips, internet, projector, restroom access, safety rules, and pickup flow.
- Prepare kits, spare batteries, chargers, tools, labels, storage bins, first aid kit, and cleaning supplies.
- Create printed attendance sheet, emergency contact sheet, parent consent form, and photo permission form.
- Prepare day-by-day lesson slides, build instructions, challenge cards, and troubleshooting sheets.
- Run a full instructor rehearsal with every kit before students arrive.

Camp-to-Kit Conversion

Do not sell the starter kit as a box of parts. Sell it as a continuation path: students can keep building after camp, join weekly STEM Academy sessions, and prepare for advanced projects.

- Offer a camp-only starter kit discount if purchased before final showcase day.
- Include a printed quick-start guide, QR code to online lessons, and a labelled parts inventory.
- Bundle one free online group support session with the kit.

6. Stages 4-7: Kits, STEM Academy, Advanced Track, Competition Team

Starter Kit

The starter kit should support the first 6-10 projects, not every possible robotics idea. Keep it affordable, durable, labelled, and aligned to your lessons.

Kit Area	Suggested Contents	Notes
Controller	Micro:bit, Arduino-compatible board, or similar beginner board	Choose one platform first; standardization reduces support problems.
Movement	2 small motors, wheels, motor driver, chassis pieces	Use rugged parts that tolerate beginner handling.
Sensors	Ultrasonic sensor, light sensor, line sensor, basic buttons	Match each sensor to a lesson challenge.
Build tools	Screwdriver, jumper wires, breadboard, battery holder	Avoid sharp or high-risk tools for take-home kits.
Learning pack	Quick-start guide, project cards, QR code to videos, parts checklist	The guide is part of the value, not an afterthought.

STEM Academy

The STEM Academy is the brick-and-mortar center. It should launch only after you have demand signals: YouTube attention, course subscribers, camp waitlists, parent enquiries, or school partnerships.

- 1 Start with rented time in an existing safe space before committing to a permanent lease.
- 2 Offer weekly STEM clubs, exam support, robotics labs, and project mentoring.
- 3 Create membership tiers: academic support, robotics club, advanced project lab, and competition team.
- 4 Build a visible project wall showing student work, certificates, photos, and challenge progress.
- 5 Use the academy as the pickup point for printed guides, kits, and parent meetings.

Advanced Kit

- Add stronger motors, servos, sensor expansion, rechargeable power, wireless modules, IoT components, camera/vision options, and advanced project briefs.
- Require completion of the starter pathway before selling the advanced kit.
- Pair every advanced kit with mentorship sessions so students do not become stuck with unused parts.

Competition Team

Team System	How to Run It
Selection	Use attendance, attitude, build quality, teamwork, and problem-solving challenges.
Training	Weekly engineering practice, coding drills, design review, documentation, and presentation coaching.
Roles	Builder, programmer, researcher, documenter, driver/operator, presenter, team captain.
Funding	Parent contributions, sponsor packages, school partnerships, showcases, and kit revenue.

Team System	How to Run It
Publicity	Publish build logs, highlight student stories, and turn competitions into YouTube recruitment content.

7. Operating Model

Weekly Operating Rhythm

Day	Leadership Focus	Teaching/Content Focus
Monday	Review enrolments, payments, enquiries, and class attendance	Plan lesson objectives and quiz updates
Tuesday	Record or edit YouTube/tutorial content	Respond to student questions
Wednesday	Live revision class or physical class	Update course pages and assessment items
Thursday	Parent communication and sales follow-up	Prepare robotics/camp materials
Friday	Finance, stock, kit checks, and marketing review	Publish weekly content and email
Saturday	Physical classes, workshops, camps, or STEM Academy sessions	Capture photos/video evidence with permission
Sunday	Rest, strategy, and next-week planning	Light admin only

Roles to Add as Revenue Grows

- Founder/academic lead: curriculum, teaching quality, parent trust, partnerships.
- Teaching assistant: attendance, quizzes, student support, class setup.
- Content assistant: editing, thumbnails, uploads, worksheets, social snippets.
- Operations assistant: payments, forms, kit inventory, parent messaging.
- Robotics mentor: camp delivery, project support, competition team coaching.

Facility Requirements for Physical Classes

Need	Minimum Standard
Safety	Visible entrance, emergency contacts, first aid kit, child pickup controls, clear behaviour rules
Learning space	Tables for teams, projector/screen, whiteboard, storage, reliable power, ventilation
Robotics lab	Labelled bins, charging area, tool check-out, spare parts, soldering only with controls if used
Administration	Attendance register, payment records, parent communication log, incident log
Brand trust	Clean signage, student work display, certificate wall, parent waiting/meeting area if possible

8. Pricing, Revenue, and Marketing

Revenue Streams

- CSEC subscriptions: recurring monthly or term-based plans.
- Live revision classes: included in premium subscription or sold as revision blocks.
- Printed study guides: add-on purchase, pickup or delivery.
- Physical classes: weekly or term packages.
- Robotics camps: premium holiday or weekend programs.
- Starter and advanced kits: product revenue plus learning continuity.
- Competition team: training fees, sponsorship, showcases, and grants where available.

Pricing Method

- 1 Calculate direct cost: platform fees, printing, venue, materials, instructor time, payment fees, and support time.
- 2 Add operating margin: marketing, admin, equipment replacement, and future development.
- 3 Compare local alternatives: tutors, revision schools, camps, and STEM clubs.
- 4 Offer simple tiers: Free, Standard, Premium, Camp/Kit add-ons.
- 5 Avoid too many discounts. Use scholarships or sponsor-funded seats instead of weakening the brand.

Marketing Funnel

Funnel Step	Asset	Success Metric
Awareness	YouTube tutorials, shorts, school/community posts	Views, subscribers, website clicks
Lead capture	Free lesson, study guide, quiz, camp waitlist	Email signups, parent enquiries
Conversion	Subscription page, parent call, limited revision cohort	Paid students, trial-to-paid rate
Retention	Weekly class rhythm, progress reports, badges, live help	Attendance, renewals, quiz completion
Expansion	Robotics camp, kits, physical classes	Upsells, referrals, testimonials
Reputation	Showcases, competition team, results stories	Sponsors, school partners, press/social proof

9. Digital Tools and AI Workflow

Use AI tools to speed up preparation, not to replace teaching judgment. Every AI-generated quiz, summary, study guide, or answer explanation must be reviewed by the academic lead before students see it.

Tool Area	Recommended Use	Control Rule
NotebookLM	Convert syllabus notes, lesson transcripts, and study guides into student-friendly summaries and Q&A; support	Use only reviewed source documents; do not upload confidential student records.
Google Gemini	Draft quiz explanations, alternate examples, parent email drafts, and revision prompts	Check facts, marks schemes, and syllabus alignment before publishing.
Website	Public landing pages, free lessons, subject navigation, enquiries	Do not store private student or payment data in public code.
Course platform	Paid lessons, quizzes, completion tracking, user access	Use proper authentication and role controls.
Video host	YouTube for free content, protected video host for paid content	Unlisted YouTube links are not strong access control.
CRM/email	Parent follow-up, onboarding, camp reminders, renewal notices	Keep messages professional and permission-based.

AI quality gate: check syllabus alignment, scientific accuracy, age appropriateness, answer key correctness, and tone before publishing any AI-assisted material.

10. Step-by-Step Launch Roadmap

First 30 Days

- 1 Finalize brand promise, logo use, website home page, and first subject pathway.
- 2 Publish five YouTube videos and link each one to a matching website page.
- 3 Create one free study guide and one free quiz for lead capture.
- 4 Set up enquiry form, email address, parent contact workflow, and spreadsheet/CRM tracking.
- 5 Draft pricing and rules for the first CSEC subscription offer.
- 6 Create consent, attendance, refund, privacy, and parent communication templates.

Days 31-60

- 1 Launch the paid CSEC subject to a small founding group.
- 2 Run one live revision class and collect feedback.
- 3 Publish weekly YouTube content and update lessons based on questions asked.
- 4 Start a robotics camp waiting list and survey parents on dates, ages, and price tolerance.
- 5 Create the first starter kit bill of materials and test one kit fully.

Days 61-90

- 1 Improve the subscription course based on student usage and quiz results.
- 2 Confirm venue and run a small robotics camp or workshop pilot.
- 3 Photograph student projects with consent and create a public showcase page.
- 4 Offer starter kits to camp participants and subscribers.
- 5 Prepare expansion decision: add second CSEC subject, repeat camp, or start weekly STEM club.

Months 4-12

Month	Milestone
4	Second subject or deeper Biology content; first structured parent progress update
5	Second robotics workshop; starter kit refinement and supplier shortlist
6	Term-based STEM club pilot; document student projects and testimonials
7	Assess physical academy demand and venue options
8	Launch advanced projects for strongest students
9	Prepare sponsorship deck and school/community partnership offer
10	Select competition team candidates and begin training
11	Host student showcase or mini-expo
12	Review KPIs, renew strategy, decide next subjects and academy expansion

11. Working Checklists

Paid Course Launch Checklist

- Course page published with clear promise, included features, price, and payment instructions.
- At least one complete section available with video, notes, quiz, and answer explanations.
- Subscriber welcome email written and tested.
- Live class link, calendar, and rules prepared.
- Support workflow ready: how students ask questions and when they receive answers.
- Refund/cancellation policy written in plain language.

Robotics Camp Checklist

- Venue confirmed in writing; power, seating, internet, restroom, and emergency plan checked.
- Student registration, consent, photo permission, pickup list, and emergency contacts collected.
- Kits tested, spare parts packed, batteries charged, tools counted and labelled.
- Instructor script, slides, project cards, certificates, and final showcase plan ready.
- Parent reminder sent 48 hours before camp with arrival, pickup, food, and dress instructions.

Monthly Review Checklist

Metric	Question to Ask
Audience	Are YouTube views and website visits increasing?
Conversion	How many free users became paying subscribers?
Learning	Are quiz scores, attendance, and completion improving?
Retention	Are students renewing and attending live classes?
Operations	Which tasks are taking too much founder time?
Finance	Which offer has the best margin and repeat demand?
Safety	Were there incidents, near misses, or parent concerns?

12. Risk Controls

Risk	Control
Overbuilding too early	Launch one subject and one camp pilot before hiring heavily or leasing permanent space.
Weak student outcomes	Use quizzes, live revision, parent updates, and targeted intervention lists.
Video piracy or link sharing	Keep free videos on YouTube; use protected hosting/platform access for paid core lessons.
Child safety incident	Use consent forms, attendance logs, pickup controls, supervision ratios, and incident reports.
Inventory losses	Label kits, count tools before/after sessions, keep spare parts, and charge deposits where appropriate.
Founder overload	Document SOPs early and hire assistants for repeatable admin/content tasks.
AI mistakes	Review every AI-assisted explanation, quiz, answer key, and study note before publishing.

Important: confirm legal, tax, child safety, data privacy, employment, insurance, and venue requirements with qualified local professionals before operating paid physical programs.

13. Templates

Lesson Page Template

- Title and syllabus objective.
- Short student-friendly outcome: By the end, you can...
- Embedded video.
- Concise study guide.
- Flashcards or key terms.
- Self-assessment quiz with feedback.
- Next lesson link and paid upgrade prompt where appropriate.

Parent Enquiry Script

Thank you for contacting EduTech Academy. To recommend the right support, please share the student's form level, subject, current challenge, exam timeline, and whether you prefer online revision, physical class, robotics/STEM program, or a blended plan.

Student Progress Note

Field	Prompt
Student	Name, subject, cohort
Attendance	Sessions attended / missed
Strengths	Concepts or skills improving
Support needed	Topics to revisit or practice
Next action	Assigned video, quiz, class, or parent follow-up

End of manual. Use this document as a living operating guide. Update it after every pilot, camp, term, and major launch decision.